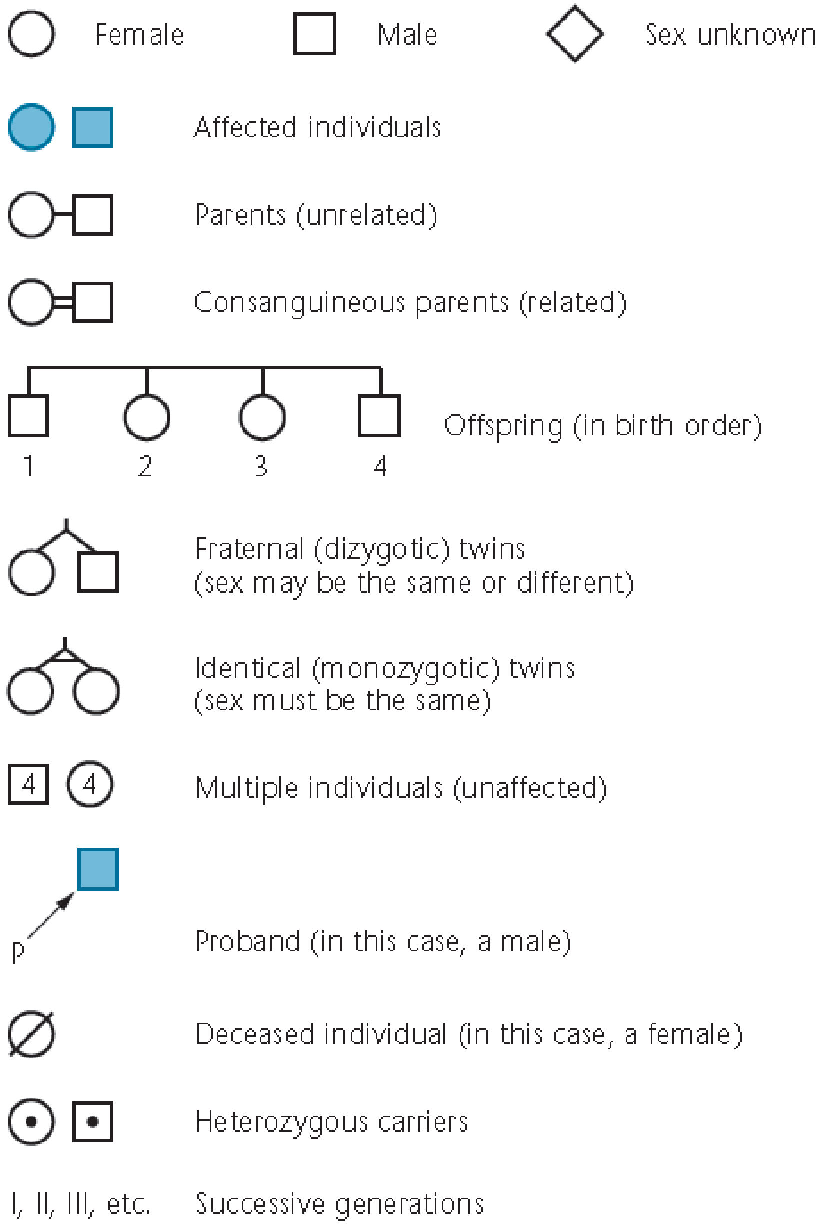
Pedegree analysis

Crossing/breeding is not done in humans,

So, we check inheritance patterns of traits (often genetic diseases) running in families



consanguineous—such as first cousins, they are connected by a double line.

Offspring are called **sibs** (short for **siblings**) and are connected by a horizontal **sibship line.**

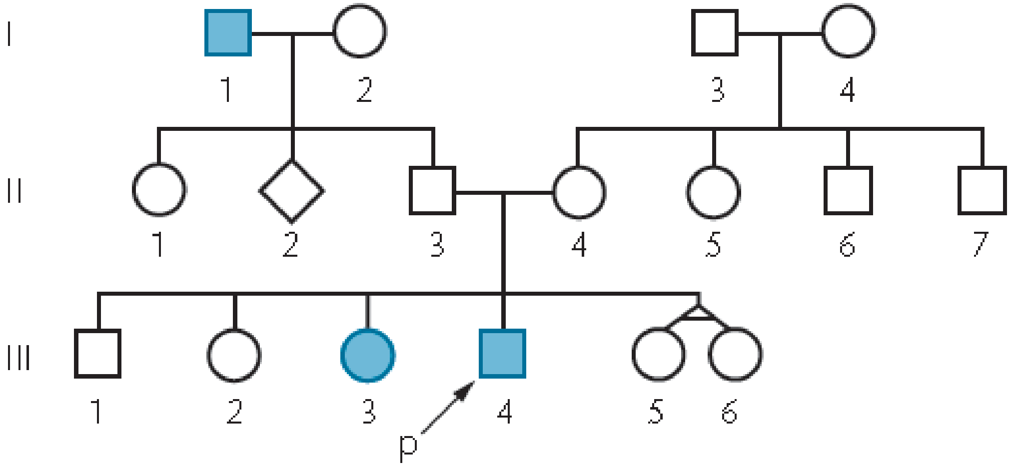
Sibs are placed in birth order from left to right and are labeled with Indo-Arabic numerals.

Parents also receive an Indo-Arabic number designation.

Each generation is indicated by a Roman numeral.

Autosomes – all 2*22 Chrs
Sex chromosomes – X,Y

(a) Autosomal Recessive Trait

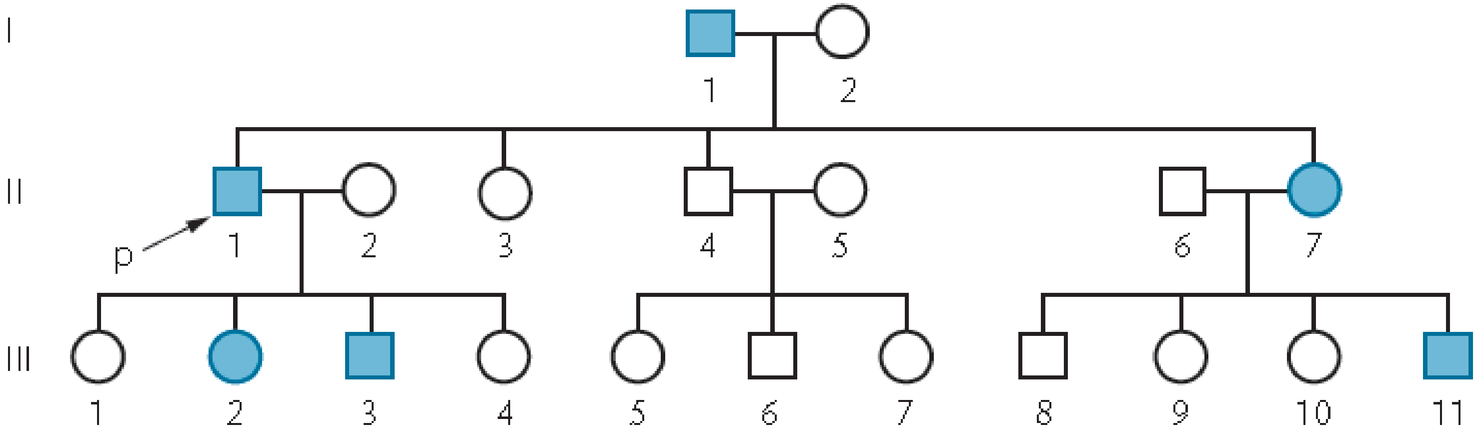


Either I-3 or I-4 must be heterozygous

Recessive traits typically skip generations

Recessive autosomal traits appear equally in both sexes

(b) Autosomal Dominant Trait

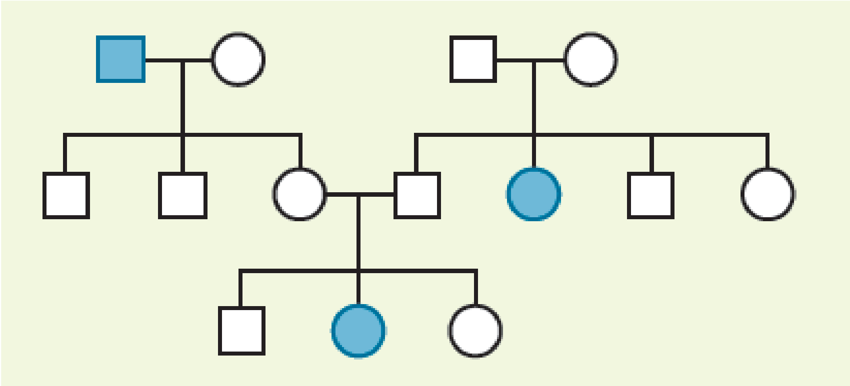


I-1 is heterozygous for a dominant allele

Dominant traits almost always appear in each generation

Affected individuals all have an affected parent. Dominant autosomal traits appear equally in both sexes

A problem



The following pedigree is for myopia (nearsightedness) in humans. Predict whether the disorder is inherited as the result of a dominant or recessive trait. Determine the most probable genotype for each individual based on your prediction

The key to its solution is to identify whether or not there are individuals who express the trait but neither of whose parents also express the trait. Such an observation is a powerful clue and allows you to rule out one mode of inheritance.